Mukesh Patel School of Technology
Management & Engineering

LinguaRoute

Language identification for multilingual content routing

Natural Language Processing (702AI0C009)

I081 Dhruv Paleja | I053 Manan Shah

Faculty: Dr. Ami Munshi

1. Problem statement

A multilingual team receives text before it knows which language desk should handle it. Checking every paragraph manually is repetitive. A language suggestion can help a reviewer organize incoming content.

2. Objectives

- Identify 18 supported written languages.
- Show the nearest alternatives and useful character patterns.
- Inspect language changes between sentences and export batch suggestions.

3. Proposed approach

Supervised character TF-IDF and a linear support vector machine trained for this project. Multinomial Naive Bayes is the comparison baseline. No pretrained service or large language model is used.

4. Data and evaluation design

WiLI-2018, created by Martin Thoma, supplies labelled Wikipedia paragraphs. Eighteen languages are selected, with 500 original training and 500 test paragraphs per language. Several languages share a script, including Hindi/Marathi and the selected Latin-script languages.

Exact normalized training/test overlaps and repeated training rows are removed. The cleaned training data is split 80/20 with stratification for validation. Vocabulary, weights and hyperparameter choices do not use test labels. Near-duplicates and article/topic overlap are not fully removed.

5. Methodology

Stage	Method	Purpose
Normalize	NFC Unicode, lowercase, spaces	Preserve accents and script
Represent	Character 2-4 grams; TF-IDF	Capture recurring spelling patterns
Classify	Linear SVM; 18 language scores	Choose the leading candidate
Review	Length, score gap and coverage	Flag weak suggestions
Use	Sentence map; batch CSV	Support a human language desk

Text normalization, character n-grams, TF-IDF, supervised classification, validation and precision/recall/F1. Language identification here uses orthographic patterns, not full semantic understanding.

6. Application example

Input: Our team would like more information about the international conference.
Output: English is the leading language suggestion. The app shows alternative languages, character patterns and a review status.

7. Results

Measure	Naive Bayes	Linear SVM
Accuracy	97.54%	98.80%
Macro F1	0.9765	0.9881

9,000 held-out paragraphs; 18 languages. Clean training set: 8,975. These are paragraph benchmark results, not chat or unknown-language accuracy.

Interpretation

The SVM performs better than the Naive Bayes baseline on the same test paragraphs. Character patterns can separate languages that share a script. English has the lowest per-language F1 (0.9412), which shows why the average alone is not enough.

User workflow

Enter text or choose an example, inspect the suggested fields or result, review the interpretation, then export the desired output. Computation runs in the browser; no paid inference API is required.

8. Limitations

- Only 18 known languages are supported; unknown languages can still be assigned a label.
- Wikipedia results do not establish performance on noisy chats or romanised Hindi.
- The sentence map classifies segments separately; it is not token-level code-switch detection.
- Short messages, shared words and mixed-language sentences can be ambiguous.

9. Conclusion and future work

A compact language classifier can support a practical routing workflow while showing its alternatives and limitations. The next useful improvement is a separate evaluation on consented short messages and out-of-scope languages.

References

1. Thoma (2018), WiLI benchmark dataset. https://arxiv.org/abs/1801.07779
2. WiLI-2018 dataset, ODbL. https://doi.org/10.5281/zenodo.841984
3. scikit-learn: LinearSVC. https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html
4. Live demo: linguaroute-dhruv.vercel.app. https://linguaroute-dhruv.vercel.app
5. Source code: github.com/dhruvpaleja/linguaroute-nlp. https://github.com/dhruvpaleja/linguaroute-nlp